No useful parameter regime for the proposed QC-GHP stopping bound

24 October 2026 · Universal negative answer · L-F16-stopping-bound

Abstract. We prove that no positive-dimensional binary biregular parameter choice in the scalar-factor QC-GHP model used by Abiad and Castriota's Proposition 5.1 yields a stopping-distance guarantee exceeding two. With at least two row blocks, an unavoidable scalar-block spectral contribution contradicts the required inequality. With one row block, the inequality forces density above three quarters and a large root in (1,2]. Positive column weight excludes the small-size branch, but already gives the same integer lower bound. The raw spectral inequality can hold: a checked dense CSS example attains it with stopping distance two. This is a universal impossibility proof for useful parameter choices, not a failed search or a counterexample to the proposition's inequality. All proofs and matrix/root translations are checked in Lean.

Question, source and answer

Question / source	Answer / evidence
Original parameter question: Abiad-Castriota2026, Prop.5.1 / Remark5.2 / §6	Answered negatively in its stated model: no useful root above2. Universal checked nonexistence theorem.
Stronger replacement-bound direction in the same §6	Still open; distinct from the answered parameter question.

Documentation revision: full Lean URLs, actual proof excerpts, dependency guide and complete calendar-delta ledger are supplied in Appendices A-C. No existing proof or source-question count changes; the additional dense-source bridge is explicitly identified.

1. Source question and precise negative answer

Aida Abiad and Nichola Castriota [1], 15 July 2026, arXiv:2607.13666v1, Remark 5.2 and §6, ask for QC-GHP design parameters making Proposition 5.1 informative, and separately for a stronger spectral bound. We answer the useful-parameter component negatively for the construction stated in that paper. Its §3.2 specializes to $r_2=n_2=1$, describing this as "used throughout this paper"; Proposition 5.1 uses Theorem 4.1's corresponding scalar-shift Fourier matrices. General two-matrix or nonabelian products are extensions beyond this proposition's setup, rather than missing parameter cases in the theorem below.

Let r,n_1,ℓ,γ be positive integers, k a natural number, and $m=r\ell\geq 2$. Let A be the expanded $r\ell$-by-$n_1\ell$ first block and B the ℓ-by-ℓ scalar circulant. Assume the real expanded matrix $H=[A \mid I_r\otimes B^T]$ has entries zero or one, column weight γ and row weight k. Put $s=\sigma_2(H)^2$, counting Gram eigenvalues in descending order with multiplicity. Define the original source roots by

$$a_\pm=(m/\gamma)(1\pm\sqrt{(1-2s/\gamma)}).$$

Theorem 1 (universal obstruction). There are no such parameters and matrices satisfying both $s<\gamma/2$ and $a_+>2$. More precisely, if $s<\gamma/2$ then $r=1$ and $1<a_+\leq 2$. Every nonempty stopping support S satisfies $|S|\geq 2$, $a_-<|S|$ and $a_+\leq|S|$. Thus the small-size branch is excluded, while the resulting integer guarantee is exactly two and follows already from positive column weight.

All positive row-block counts and both finite and asymptotic parameter choices are covered. Circulancy, CSS commutation, girth and connectivity are unnecessary for the obstruction: the formal theorem proves it for a larger matrix class. Positive dimensions, positive column degree and $m\geq 2$ are explicit hypotheses. Empty first factors, $\gamma=0$ and matrices without a second singular value are outside this statement. No logical-distance claim is made.

2. Proof of the universal obstruction

Case r≥2. At zero Fourier frequency the repeated scalar block contributes $\gamma^2 I_r$ to a positive semidefinite Gram matrix. At least two eigenvalues of this block are at least γ^2; removing one top eigenvalue leaves $s \geq \gamma^2$. For positive integral γ this contradicts $s < \gamma/2$. The Lean proof works directly with the expanded matrix: on vectors constant within row fibers the scalar block supplies energy γ^2 times squared norm, and the first block adds a nonnegative quantity. In a two-dimensional subspace, a nonzero vector orthogonal to a top Gram eigenvector has energy at most s times squared norm. This also covers repeated top eigenvalues.

Case r=1. Now $m=\ell$ and $N=qm$ with $q=n_1+1\geq 2$. Edge counting gives $k=q\gamma$, and binary column regularity gives $\gamma \leq m$. The trace and a weighted Cauchy–Schwarz bound give

$$k(m-\gamma) \leq (m-1)s. \quad (T)$$

Indeed $\operatorname{tr}(HH^T)=mk$, its largest eigenvalue is at most γk, and every other eigenvalue is at most s. Subtracting the largest from the trace proves (T). Combining it with the strict gap yields $\gamma > m-(m-1)/(2q) \geq (3m+1)/4$, so the surviving matrices have column density above three quarters.

To bound the root, multiply (T) by γ. Since $q\geq 2$ and $m^2 \geq \gamma(m-1)$, and since $s\geq 0$,

$$sm^2 \geq 2\gamma^2(m-\gamma),$$
$$1-2s/\gamma \leq (2\gamma/m-1)^2.$$

The density estimate makes $2\gamma/m-1$ nonnegative. Taking nonnegative square roots proves $a_+ \leq 2$. Strict positivity of the discriminant and $m/\gamma \geq 1$ imply $a_+ > 1$. The same hypotheses give $0 \leq a_- < 2$. A singleton support cannot be stopping because its positive column contains a neighboring check of induced degree one. Hence $|S|\geq 2$ excludes the entire small-size branch and already implies $|S| \geq a_+$. These two cases exhaust every positive r. □

3. Raw inequality and the separate branch requirement

Proposition 5.1 gives $a(m-a\gamma/2) \leq sm^2/\gamma^2$. In the strict-gap regime it yields the disjunction $a \leq a_-$ or $a \geq a_+$; exclusion of the small branch is an additional requirement. Theorem 1 explicitly checks that exclusion and then shows the large-root conclusion adds nothing beyond $|S|\geq 2$. Increasing column weight or choosing better constituent expanders cannot change this conclusion within the stated model.

The previously implicit source identification is now checked explicitly: A and B are the length-two circulants with support {0,1}; $H_X=[A|B^T]$ and companion $H_Z=[B|A^T]$ become the displayed matrices under proved row/column bijections. DenseSourceBridge.source_certificate combines the actual degrees, spectrum, roots, CSS and stopping statements.

Checked example. The all-ones 2-by-4 pair $H_X=H_Z$ is a scalar-factor CSS example with $\gamma=2$, $k=4$, $s=0$, $a_-=0$ and $a_+=2$. Its stopping distance is two and $H_X H_Z^T=0$ over F_2. Thus asserting that the raw spectral inequality never holds would be false. The proved nonexistence concerns a useful improvement beyond two. This does not refute the proposition’s inequality.

4. Checked statements and proof coverage

The new theorem below combines the previously checked row-block cases. The matrix, degree, stopping and original-radical bridges are proved, not assumed as spectral-envelope hypotheses. Here eigs H 1 means the second ordered eigenvalue of the actual real HH^T.

no_parameters_give_source_improvement :
∀ positive r, γ and positive finite dimensions,
∀ binary (γ,k)-biregular H=[A | $I_r \otimes B^T$], m≥2,
¬(eigs H 1<γ/2 ∧ 2<sourceLargeRoot m γ (eigs H 1)).

The companion theorem source_parameter_question_negative also concludes r=1, $1<a_+\leq 2$ and, for EVERY nonempty stopping support, $|S|\geq 2$, $a_-<|S|$ and $a_+\leq|S|$. The exact Lean signatures, including nonempty first-factor and fiber index types, appear in the axiom audit.

Proof or bridge	Checked declaration / source
Actual real Gram spectrum, trace and energy	Spectral.lean: energy_eq_gram, energy_eigenbasis, energy_le_of_coeff_zero, eigs_trace
Binary biregular trace bound and dimensions	Spectral.lean: binary_weighted_cauchy, regular_trace_obstruction, binary_column_sum_le_card, regular_edge_count, one_row_cardinal_obstruction
Every r≥2: scalar block obstruction	Spectral.lean: scalarFactor_energy_lower, replicated_scalar_second_lower, qc_scalar_factor_no_source_regime
Original radical translation and r=1 roots	Spectral.lean: sourceLargeRoot_eq, sourceSmallRoot_eq, one_row_density, one_row_large_root_le_two, one_row_roots
Stopping definition and small branch exclusion	Spectral.lean: IsStopping, stopping_card_ge_two, scalar_one_row_source_obstruction
Dense source construction, CSS and exact stopping distance	DenseSourceBridge.lean: source_certificate; Spectral.lean: denseExample_second, denseExample_css, denseExample_source_certificate, denseExample_minimum
All positive r; direct nonexistence statement	SourceQuestionAnswer.lean: source_parameter_question_negative, no_parameters_give_source_improvement

The complete proofs have immutable repository links: SourceQuestionAnswer.lean and Spectral.lean. The checked signatures and axiom audit are pinned to the same source revision, b7628643bb25. PROOF-LINKS.html supplies additional file links. Hashes in verification.json identify the checked files and this publication revision. The companion X text remains an unposted draft.

Pinned environment: Lean 4.36.0-rc1; mathlib 5f7b4198d22a9c82e1c698a379334fe128095c82. The aggregate build succeeded (5,470 jobs). Both new theorems use only propext, Classical.choice and Quot.sound. The inherited Spectral.lean is unchanged, with SHA-256 a3e712b2a206bd7743408ecc122e66e7d980dbd4689c6d49977536dfe9bb469d; its 52-declaration audit is preserved. No admitted lemma, native numerical oracle or sampled experiment is a premise.

5. Prior art, chronology and what remains open

The parent investigation compared the exact source with Tanner’s bit-oriented spectral argument [2], Zhu–Chugg’s regular-graph stopping bound [3], the original lifted-product construction [4], and quantum Tanner constructions [5]. Zhu–Chugg’s expression $N(2\gamma-s)/(\gamma k-s)$ uses a different threshold and has no small/large-support branch; it does not supply parameters making the source test useful. Constituent Ramanujan expansion is not a certificate for the expanded H_X singular-value condition. The parent prior-art ledger includes inspected hypotheses, public GitHub code and accessible X posts, failed searches and full-text gaps. No global absence or priority claim follows from those bounded searches. The impossibility result follows from the proof above.

Separate checked follow-ups establish stopping distance six for the B1 positive 63-multiple cyclic-cover family and classify the base matrix’s 441 size-six supports as unique companion Z-check rows. Those completed results use incidence information; they do not make Proposition 5.1 informative. They remain in the preceding incidence and classification artifacts. No additional paper or source question was encountered during this scope clarification.

Source / milestone	Date and precision
Tanner spectral ingredient [2]	Inspected manuscript 26 July 2000; journal 2001
Zhu–Chugg [3]	Conference 28 September 2005; inspected journal December 2007 (month precision)
Panteleev–Kalachev [4]	First preprint 4 April 2019; inspected v3 12 November 2021
Leverrier–Zémor [5]	First preprint 28 February 2022; inspected v3 16 September 2022
Abiad–Castriota question [1]	15 July 2026, v1
This revision	24 October 2026; 101 calendar days after [1]

Investigation began 24 October at 19:49 UTC; the original obstruction audit completed at 20:20:37 UTC. The all-cases theorem and corrected source-scope explanation were assembled after 22:20 UTC. Exact build/audit times are recorded in verification.json. The 101-day source-to-revision difference is date arithmetic, not a claim that a problem remained unsolved for that duration.

Question answered. Can parameters in the paper’s stated positive-dimensional binary biregular scalar-factor QC-GHP model make Proposition 5.1 improve the stopping-distance guarantee beyond two? No: Theorem 1 proves nonexistence for every such parameter choice. The shared inventory records this as the fifth completed canonical Lean target (5/135); earlier checkpoint manifests retain their dated four-target counts. If “informative” means only the raw strict spectral inequality, the dense example instead gives a positive answer; it supplies no useful improvement. **Question still open.** A generally useful stronger replacement or tightening of the bound. The original canonical parameter question is marked answered negatively; the stronger-bound direction remains separately open. **Next proposed target (follow-up).** Prove an incidence-conditioned replacement that improves on two on a sparse family beyond the already completed B1 results.

References

[1] A. Abiad and N. Castriota. Spectral and Additive Combinatorial Methods for Cycles and Absorbing Sets in Lifted-Product Quantum LDPC Codes. arXiv:2607.13666v1, 15 July 2026, §3.2, Proposition 5.1, Remark 5.2, §6. https://arxiv.org/html/2607.13666v1

[2] R. M. Tanner. Minimum-Distance Bounds by Graph Analysis. Manuscript 26 July 2000; IEEE Transactions on Information Theory 47(2), 808–821 (2001).

[3] M. Zhu and K. M. Chugg. Bounds on the Expansion Properties of Tanner Graphs. IEEE Transactions on Information Theory 53(12), 4834–4840 (December 2007), Corollary 6. https://hal.usc.edu/chugg/docs/pubs/zhu2007bounds.pdf

[4] P. Panteleev and G. Kalachev. arXiv:1904.02703v3, inspected 12 November 2021. https://arxiv.org/abs/1904.02703v3

[5] A. Leverrier and G. Zémor. Quantum Tanner codes. arXiv:2202.13641v3, 16 September 2022. https://arxiv.org/abs/2202.13641v3

Appendix A. Actual checked Lean and dependencies

This is a documentation revision of an already checked result, not a new research target. The declaration and proof below are extracted verbatim from the committed Lean source. The earlier proof-to-Lean tables identify the individual mathematical claims; Appendix B gives full current and immutable URLs for every research and audit module in this artifact, including inherited dependencies.

```
theorem no_parameters_give_source_improvement
    {C F : Type*} [Fintype C] [DecidableEq C] [Nonempty C]
    [Fintype F] [DecidableEq F] [Nonempty F]
    (r γ k : ℕ) (hr : 0 < r) (hγ : 0 < γ)
    (A : Matrix (Fin r × F) (C × F) ℝ) (B : Matrix F F ℝ)
    (hm : 2 ≤ Fintype.card (Fin r × F))
    (hbin : ∀ i j, scalarFactorMatrix A B i j = 0 ∨ scalarFactorMatrix A B i j = 1)
    (hcol : ∀ j, ∑ i, scalarFactorMatrix A B i j = (γ : ℝ))
    (hrow : ∀ i, ∑ j, scalarFactorMatrix A B i j = (k : ℝ)) :
    ¬ (eigs (scalarFactorMatrix A B) ⟨1,hm⟩ < (γ : ℝ)/2 ∧
      2 < sourceLargeRoot (Fintype.card (Fin r × F)) γ
        (eigs (scalarFactorMatrix A B) ⟨1,hm⟩)) := by
  rintro ⟨hgap,huseful⟩
  have h := source_parameter_question_negative r γ k hr hγ A B hm hbin hcol hrow hgap
  exact (not_lt_of_ge h.2.2.1) huseful
```

Dependency guide

mathlib ordered Gram spectrum -> Spectral energy/trace -> multi-row obstruction and one-row roots -> SourceQuestionAnswer universal nonexistence theorem. AuditSourceAnswer checks its axiom closure; the parent Audit checks the52 inherited declarations. DenseSourceBridge separately constructs and reindexes the actual length-two scalar-circulant HX/HZ example and combines its degrees, spectrum, roots, CSS and stopping contract; AuditDenseSourceBridge checks all20 new declarations.

Translation and boundary check: source binary expanded incidence is the real zero-one matrix used for its Gram spectrum, with descending eigenvalues counted with multiplicity; stopping supports are nonempty and permit stabilizer supports. The source roots retain the separate small-size branch. The cover and classification interfaces instead use the actual binary HX2/HZ2 matrices; base-only classification and positive-cover distance are distinct statements. These definitions and bridges, rather than a numerically assumed spectral envelope, carry the translation.

The imported research modules are included in the published source tree. Only standard mathlib axioms appear in the recorded audits. Source interpretation and prior-art comparisons remain documentary judgments; a successful Lean build alone does not certify those judgments. The only added formal work is the explicitly identified dense-source bridge in the parameter-answer artifact; the earlier universal, family and classification proofs remain unchanged.

Appendix A (continued). Dense source-construction bridge

Newly checked consumer: the source constituents are length-two support circulants for1+x. The definitions above this theorem in DenseSourceBridge.lean construct actual HX/HZ and bijective coordinate maps. This theorem checks the complete dense example in those coordinates. All20 declarations in the module have a standard-axiom audit; no external numerical result is assumed.

```
theorem source_certificate :
    B = supportCirculant ({0,1} : Finset (ZMod 2)) ∧
    expanded = denseExample ∧ binary = denseExampleBinary ∧
    binaryZ = denseExampleBinary ∧
    (∀ j, (∑ i, expanded i j) = 2) ∧ (∀ i, (∑ j, expanded i j) = 4) ∧
    eigs expanded ⟨1,by norm_num⟩ = 0 ∧
    eigs expanded ⟨1,by norm_num⟩ < (2:ℝ)/2 ∧
    sourceSmallRoot 2 2 (eigs expanded ⟨1,by norm_num⟩) = 0 ∧
    sourceLargeRoot 2 2 (eigs expanded ⟨1,by norm_num⟩) = 2 ∧
    binary * binaryZᵀ = 0 ∧
    (∀ S, IsStopping expanded S ↔ 2 ≤ S.card) := by
  refine ⟨rfl,expanded_eq,binary_eq,binaryZ_eq,column_weight,row_weight,?_,?_,?_,?_,?_,?_⟩
  · rw [expanded_eq]; exact denseExample_second
  · rw [expanded_eq,denseExample_second]; norm_num
  · rw [expanded_eq,denseExample_second]; rw [sourceSmallRoot_eq 2 2 0 (by norm_num) (by norm_num)]; norm_num
  · rw [expanded_eq,denseExample_second]; rw [sourceLargeRoot_eq 2 2 0 (by norm_num) (by norm_num)]; norm_num
  · rw [binary_eq,binaryZ_eq]; exact denseExample_css
  · rw [expanded_eq]; exact denseExample_stopping
```

Appendix B. Complete Lean file URLs

Every URL is printed in full and is clickable. Current links follow master; immutable links identify the exact checked file at revision b7628643bb256bf9c1a319f16f6e19608116cff6. Audit programs and vendored dependencies are included rather than hidden behind folder links.

Audit.lean - axiom/signature audit

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/Audit.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/Audit.lean

Spectral.lean - checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/Spectral.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/Spectral.lean

source-answer/AuditDenseSourceBridge.lean - axiom/signature audit

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/source-answer/AuditDenseSourceBridge.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/source-answer/AuditDenseSourceBridge.lean

source-answer/AuditSourceAnswer.lean - axiom/signature audit

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/source-answer/AuditSourceAnswer.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/source-answer/AuditSourceAnswer.lean

source-answer/DenseSourceBridge.lean - checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/source-answer/DenseSourceBridge.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/source-answer/DenseSourceBridge.lean

source-answer/SourceQuestionAnswer.lean - checked proof module

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/source-answer/SourceQuestionAnswer.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/source-answer/SourceQuestionAnswer.lean

source-answer/dependencies/Spectral.lean - inherited proof dependency

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/source-answer/dependencies/Spectral.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/source-answer/dependencies/Spectral.lean

source-answer/source-ledgers/parent-Audit.lean - axiom/signature audit

Current: https://github.com/alectionary/vibe-math/blob/master/formalization/stopping-bound/source-answer/source-ledgers/parent-Audit.lean

Immutable: https://github.com/alectionary/vibe-math/blob/b7628643bb256bf9c1a319f16f6e19608116cff6/formalization/stopping-bound/source-answer/source-ledgers/parent-Audit.lean

Appendix C. Calendar deltas for all compared sources

Civil calendar days to 2026-10-24; exact dates give exact differences. Month/year precision yields inclusive minimum/maximum day ranges; dates are not invented. Date ranges are clipped to the revision because these records already existed then. Source age is not unresolved-problem duration.

This consolidated ledger covers the original parameter result and both follow-ups. Rows explicitly distinguish full-text mathematical comparisons, metadata-only or unavailable papers, discovery provenance, software snapshots and inspected X posts. Rechecking dates did not add a paper or a new full-text question screen. Earlier table dates remain valid; this table supplies the previously missing arithmetic and additional sources.

Source / role	Milestone date and elapsed calendar days	Access / precision limit
F16: Abiad-Castriota: motivating parameter question Original question; Prop.5.1 / Remark5.2 / §6	first and inspected v1: 2026-07-15; 101 days (day precision)	inspected full text
F68: Tanner: Minimum Distance Bounds by Graph Analysis Classical spectral ingredient	earliest inspected manuscript: 2000-07-26; 9586 days (day precision) journal milestone: 2001; 9063-9427 days (year precision)	inspected full text
F67: Zhu-Chugg: Bounds on the Expansion Properties of Tanner Graphs Classical stopping bound; exact Cor.6 comparison	conference predecessor metadata: 2005-09-28; 7696 days (day precision) inspected journal version: 2007-12; 6872-6902 days (month precision)	inspected full text
F69: Panteleev-Kalachev: Degenerate Quantum LDPC Codes With Good Finite Length Performance Scalar-factor construction	first v1: 2019-04-04; 2760 days (day precision) inspected v3: 2021-11-12; 1807 days (day precision) journal milestone: 2021-11-22; 1797 days (day precision)	inspected full text
F70: Leverrier-Zemor: Quantum Tanner codes Constituent Ramanujan comparison	first v1: 2022-02-28; 1699 days (day precision) inspected v3: 2022-09-16; 1499 days (day precision) PDF internal date: 2022-09-19; 1496 days (day precision)	inspected full text
F71: Feng-Li: Spectra of Hypergraphs and Applications Classical source cited by Remark5.2; not a proof premise	publication metadata: 1996-09; 10981-11010 days (month precision)	abstract only full text gap
F72: Li-Sole: Spectra of Regular Graphs and Hypergraphs and Orthogonal Polynomials Classical comparison lead; not a proof premise	publication metadata: 1996; 10889-11254 days (year precision)	partial web text archival gap
F73: Prabhu et al.: On the Efficacy of the Peeling Decoder for the Quantum Expander Code Known stopping-set mechanisms and expansion conditions	first v1: 2025-04-30; 542 days (day precision) inspected v2: 2025-05-29; 513 days (day precision)	inspected full text
F74: Wang-Kulkarni-Poor: Finding All Small Error-Prone Substructures in LDPC Codes Classical rooted narrowing and branch-and-bound	related first preprint: 2006-05-19; 7463 days (day precision) inspected current-version date: 2009-04-22; 6394 days (day precision) journal issue: 2009-05; 6355-6385 days (month precision)	inspected full text

Source / role	Milestone date and elapsed calendar days	Access / precision limit
F75: Yao-Gokduman-Pfister: Cluster Decomposition for Improved Erasure Decoding of Quantum LDPC Codes B1 simulations / peeling-cluster comparison	first and inspected v1: 2024-12-11; 682 days (day precision)	inspected full text
Connolly2024: Connolly-Londe-Leverrier-Delfosse: Fast erasure decoder for hypergraph product codes Original stabilizer-support mechanism, credited through inspected F73/F75	first v1: 2022-08-01; 1545 days (day precision) v3 metadata: 2024-08-21; 794 days (day precision) journal publication: 2024-08-27; 788 days (day precision)	metadata only full text not consulted
F76: Prabhu-Gayathri-Kumar: On the Analysis of Stopping Sets Associated with the Quantum Hypergraph Product Code Later comparison lead; no exact theorem claim	conference metadata: 2026; 0-296 days (year precision)	full text access gap
RosnesYtrehus2007: Rosnes-Ytrehus: An Algorithm to Find All Small-Size Stopping Sets of Low-Density Parity-Check Matrices Bibliography-only enumeration lead	conference year: 2007; 6872-7236 days (year precision)	bibliography only not used as theorem evidence
Morris2025: Kirsten Morris: Analysis of Graph-based Decoders for Quantum Low Density Parity Check Codes Discovery trail through existing dissertation inventory	repository source version: 2025-08; 419-449 days (month precision)	discovery record not new full text screen
qLDPCOrg/qLDPC: qLDPCOrg/qLDPC pinned snapshot Software/infrastructure snapshot, not original publication age	inspected commit: 2026-10-23; 1 days (day precision)	pinned source or infrastructure
Lcrypto/Stopping-sets-enumerator: Lcrypto/Stopping-sets-enumerator pinned snapshot Software/infrastructure snapshot, not original publication age	inspected commit: 2023-10-29; 1091 days (day precision)	pinned source or infrastructure
leanprover-community/mathlib4: leanprover-community/mathlib4 pinned snapshot Software/infrastructure snapshot, not original publication age	inspected commit: 2026-10-21; 3 days (day precision)	pinned source or infrastructure
X-source-announcement: mathCObot source-paper announcement Accessible X comparison; not a mathematical answer	post UTC date: 2026-07-16; 100 days (day precision)	post inspected in parent audit
X-earlier-result: A_Lectionary earlier absorbing-set announcement Earlier repository result; different target	post UTC date: 2026-10-24; 0 days (day precision)	post inspected in parent audit

Source / role	Milestone date and elapsed calendar days	Access / precision limit
other-code-snapshots: Other inspected GitHub mirrors/README-only code Secondary discovery/control sources; no theorem or earliest-publication claim	recorded inspection: 2026-10-24; 0 days (day precision)	inspection date only earliest dates unassessed
Lean-runtime: Lean4.36.0-rc1 runtime Pinned proof checker; release-age claim not made	recorded use: 2026-10-24; 0 days (day precision)	use date only release date unassessed

Secondary mirrors and README-only sources have inspection-date deltas only; original publication dates remain unassessed. Lean release date likewise not verified; pinned runtime use is recorded.

The earliest date in a row means the earliest relevant milestone established by this audit, not a claim of historical priority. Tanner’s manuscript reports an older1982 submission, which is not treated as a verified public1982 publication. Metadata-only and unavailable sources supply no theorem premise and no zero-question screening claim. The X paper-announcement timestamp is16July2026UTC;15July is the arXiv paper date. Source age is distinct from the duration of an unresolved question.

The clause-by-clause audit and machine-readable date evidence are published at SKILL-COMPLIANCE-AUDIT.md and compliance-data.json. Existing named full-text, citation-index and social-search access gaps remain explicitly unresolved. Neither X posting nor a historical-priority claim is made.